nothing if we think that we already know the essentials, if we assume that they were written in a book or known by the elders of the tribe. The centuries in which people had faith in what they believed were the centuries in which little new was learned. Had they trusted the knowledge of their fathers, Einstein, Newton and Copernicus would never have called things into question and would have never been able to move our knowledge forwards. If no one had raised doubts, we would be still worshipping pharaohs and thinking that the Earth is supported on the back of a giant turtle. Even our most efficacious knowledge, such as that found by Newton, may eventually turn out, as Einstein showed, to be simplistic.

Science is sometimes criticized for pretending to explain everything, for thinking that it has an answer to every question. It's a curious accusation. As every researcher working in every laboratory throughout the world knows, doing science means coming up hard against the limits of your ignorance on a daily basis – the innumerable things which you don't know, and can't do. This is quite different from claiming to know everything. We don't know which particles we might see next year at CERN, or what our next telescopes will reveal, or which equations truly describe the world; we don't know how to solve the equations we have, and sometimes we don't understand what they signify; we don't know if the beautiful theory on which we are working is right. We don't know what there is beyond the Big Bang; we don't know how a storm works, or a bacterium, or an eye – or the cells in our own bodies, or our thought processes. A scientist is someone who lives immersed in the awareness of our deep ignorance, in direct contact with our own innumerable limits, with the limits of our understanding.

But if we are certain of nothing, how can we possibly rely on what science tells us? The answer is simple. Science is not reliable because it provides certainty. It is reliable because it provides us with the best answers we have at present. Science is the most we know so far about the problems confronting us. It is precisely its openness, the fact that it constantly calls current knowledge into question, which guarantees that the answers it offers are the best so far available: if you find better answers, these new answers become science. When Einstein found better answers than Newton, he didn't question the capacity of science to give the best possible answers – on the contrary, he confirmed it.

The answers given by science, then, are not reliable because they are definitive. They are reliable because they are not definitive. They are reliable because they are the best available today. And they are the best we have because we don't consider them to be definitive, but see them as open to improvement. It's the awareness of our ignorance that gives science its reliability.